

INTRODUCTION TO DATA MINING

DV2627 – Advanced Machine Learning



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Computer Science Department

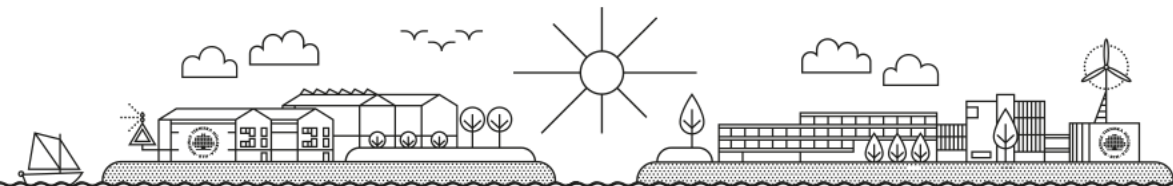
January 21, 2025



OUTLINE



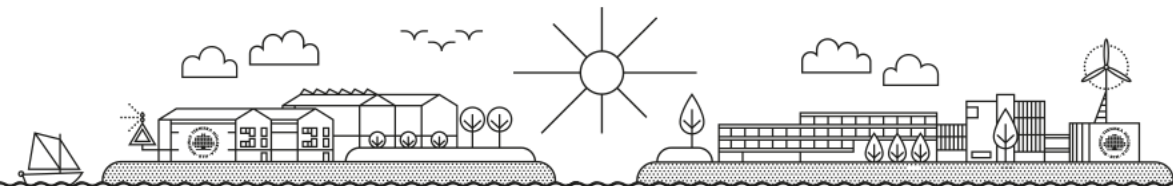
- Presentation of the course
- Presentation of the staff
- An Introduction to Data Mining



PRESENTATION OF THE COURSE

DV2627, Advanced Machine Learning, 7.5 ETCS

- 7 lectures
- 1 lab
- 3 seminars
- 2 assignments: project proposal and final project
- 3-week roll call (**Seminar 1**: Presentation of your project idea)



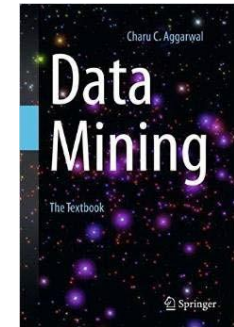
PRESENTATION OF THE COURSE (1)



Main literature:

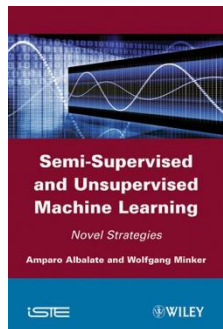
Charu C. Aggarwal. Data Mining: The Textbook. Springer, 2015.

BTH library: <https://go.exlibris.link/C8Zh3CQx>



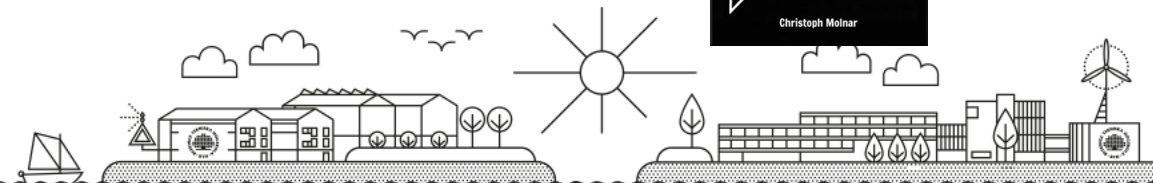
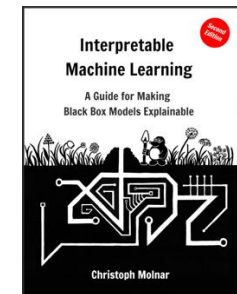
Amparo Albalade, and Minker Wolfgang. Semi-supervised and unsupervised machine learning: novel strategies. John Wiley & Sons, 2013.

BTH library: <https://go.exlibris.link/Ch7XsKQZ>



Christoph Molnar. Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.), 2024.

<https://christophm.github.io/interpretable-ml-book/>



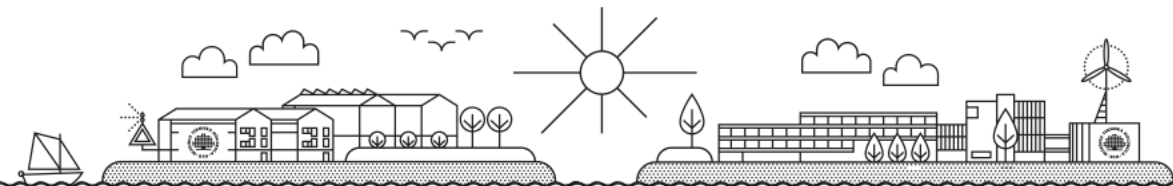
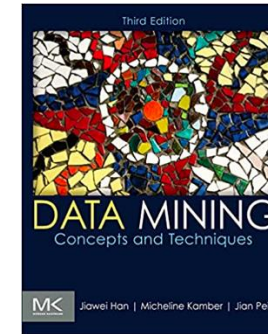
PRESENTATION OF THE COURSE (2)



Supplementary literature:

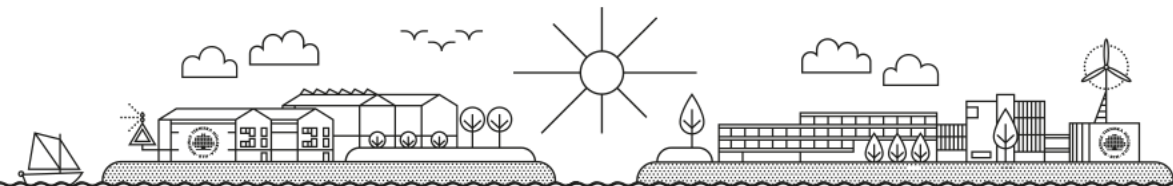
Han, Jiawei, Jian Pei, and Hanghang Tong. Data mining: concepts and techniques, 3rd edition. Morgan kaufmann, 2012.

BTH library: <https://go.exlibris.link/FXytjDD9>



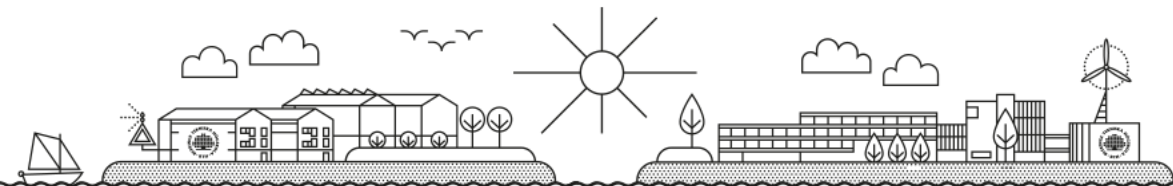
PRESENTATION OF THE STAFF

- **Main teacher:** Dr. Shahrooz Abghari, sharhooz.abghari@bth.se
- **Teacher:** Prof. Veselka Boeva, veselka.boeva@bth.se
Dr. Christian Nordahl, christian.nordahl@bth.se
- **Lecturers**
 - L1, L2, L3, L4 Shahrooz
 - L5, L6 Veselka
 - L7 Christian
- **Labs**
 - Lab Christian
- **Seminars**
 - S1, S2, S3 Shahrooz & Veselka



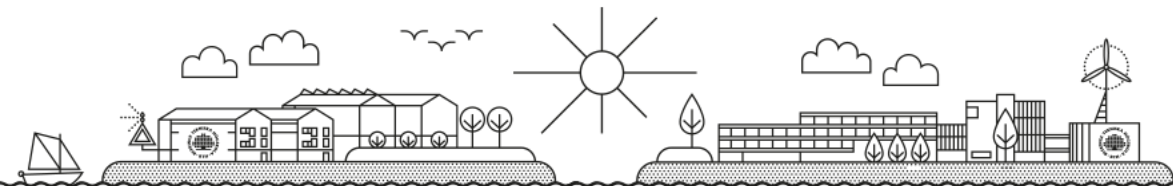
LIST OF LECTURES

1. Introduction to Data Mining
2. Data Preprocessing and Similarity and Distances
3. Association Pattern Mining
4. Outlier Detection
5. Semi-Supervised Learning
6. Mining Data Stream
7. An Introduction to XAI



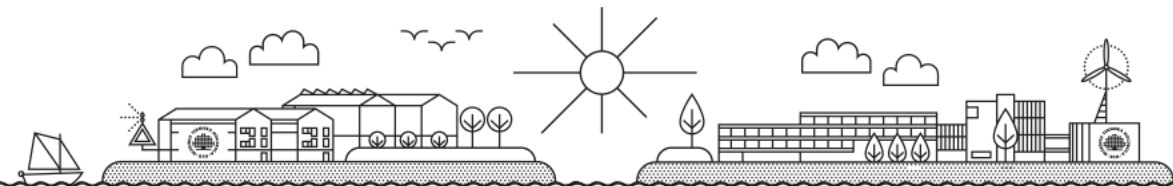
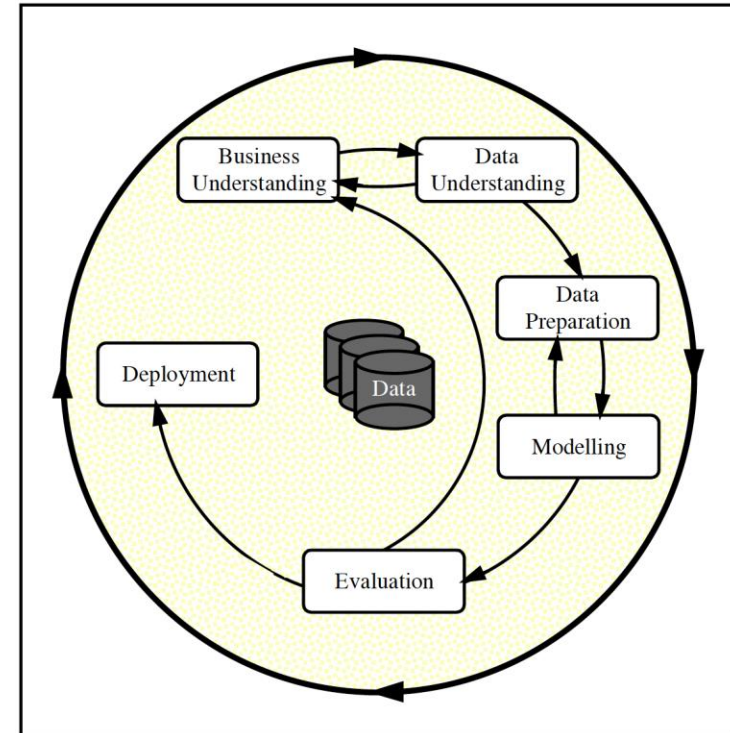
COURSEWORK

- Seminars 1 and 3, 1 ECTS, graded G/U (**Mandatory**)
 - Seminar 2, progress update
- Project proposal, 1 ECTS, graded G/U
- Final project, 5.5 ECTS graded A-F
 - Seminar 3: Final presentation (**Mandatory**)
- Total grade: 7.5 ECTS



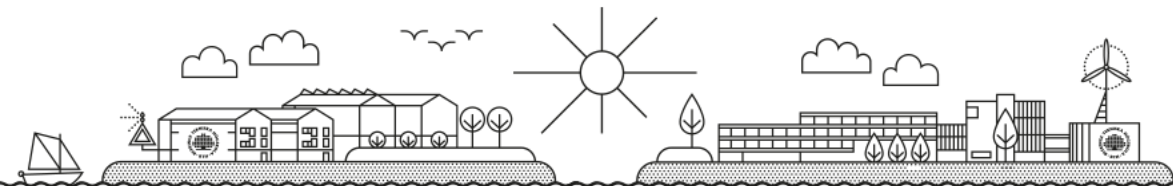
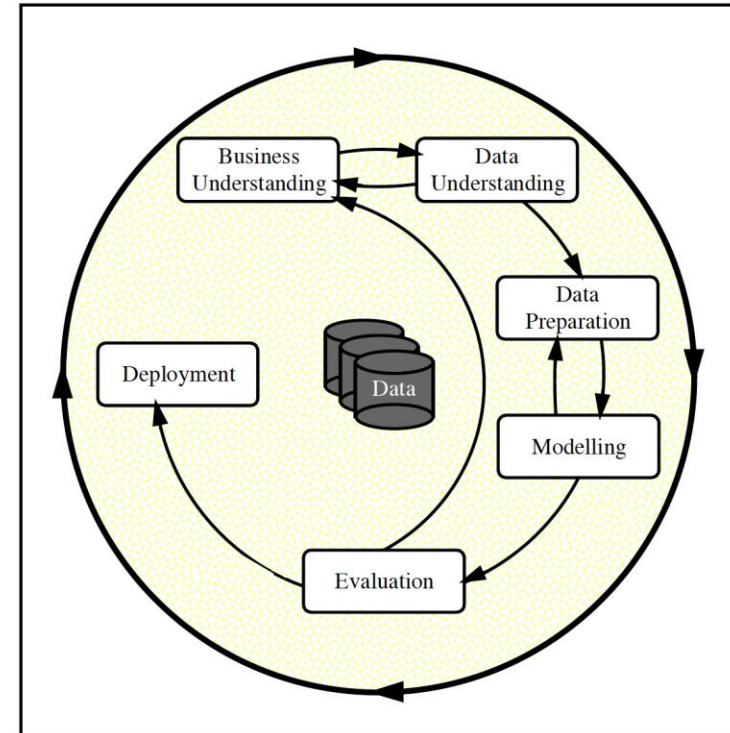
CRISP-DM

- **C**ross Industry **S**tandard **P**rocess for **D**ata **M**ining
- Aims to encourage best practices and make data mining projects:
 - cost-efficient, reliable, reproducible, manageable, and fast.



CRISP-DM (2)

1. Business understanding:
 - Focus on understanding the objectives and requirements of the project.
2. Data understanding:
 - Identify, collect, and analyze the data sets that can help you accomplish the project goals.
3. Data preparation:
 - Data munging, prepare the final data set(s) for modeling.
4. Modeling:
 - Build and assess various models based on several different modeling techniques to pick the suitable model (technical aspects).
5. Evaluation:
 - Look more broadly at which model best meets the business.
6. Deployment:
 - Generate a report and implement a repeatable data mining process.



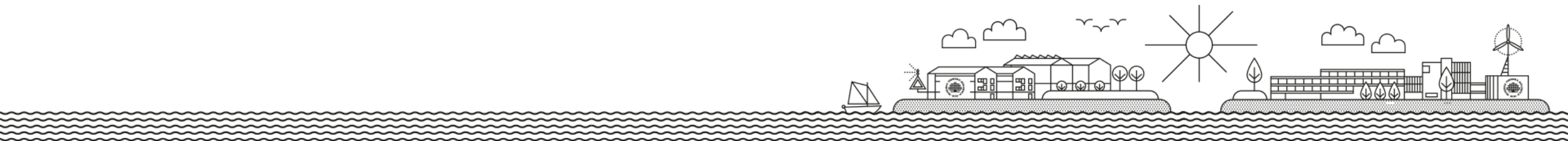
EXAMINATION



The examination is carried out as follows:

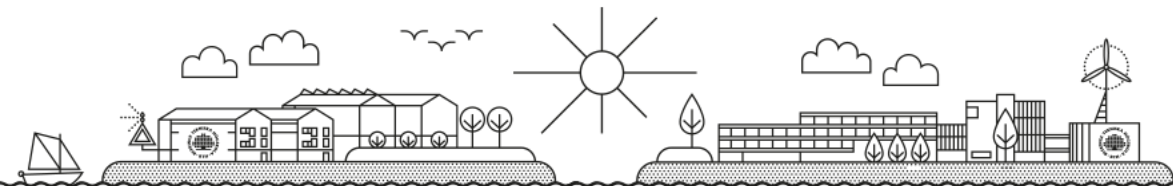
- Project Proposal: based on the written proposal report.
- Final Project: based on your presentation, implementation, and the written report.

Note: attending the seminars 1 and 3 (final presentation) are **mandatory**.



INTRODUCTION

- What is data mining?
 - “Data mining is the study of *collecting, cleaning, processing, analyzing, and gaining useful insights* from data.”
- Why do we need it?
 - Data is the new oil,
 - E.g., WWW, financial interactions, IoT devices
 - We have entered the Era of Big Data
 - Variety
 - Volumes
 - Velocity



THE DATA MINING PROCESS (1)

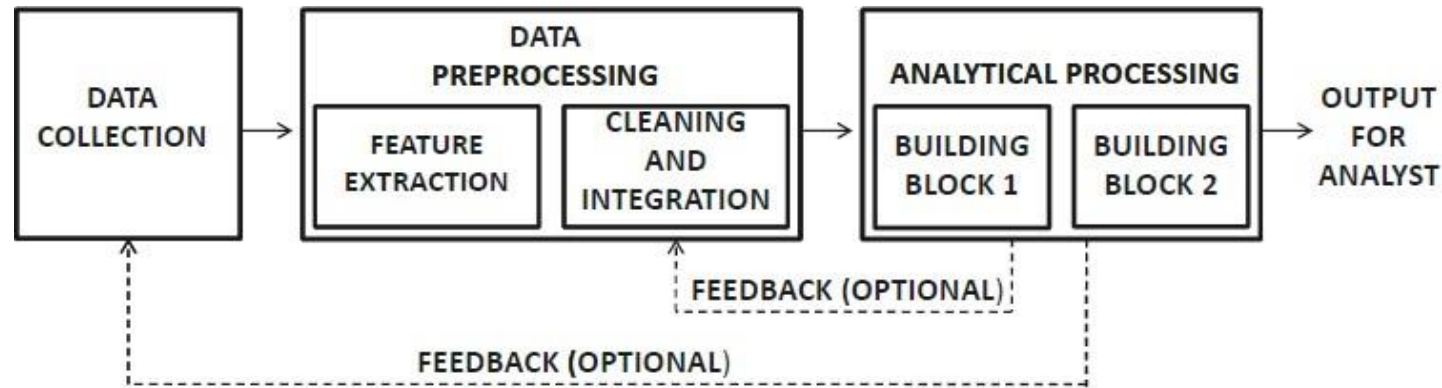
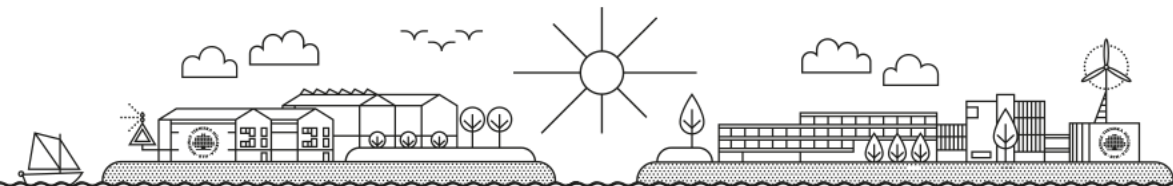


Figure 1.1: The data processing pipeline

- Data Collection
 - Hardware, Software, Human
- Feature Extraction
 - Multidimensional



THE DATA MINING PROCESS (2)

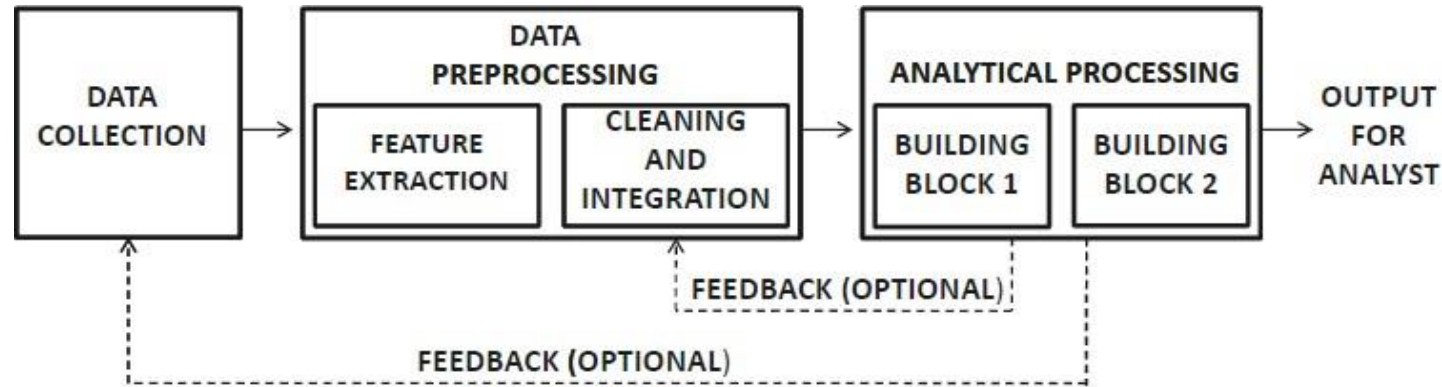
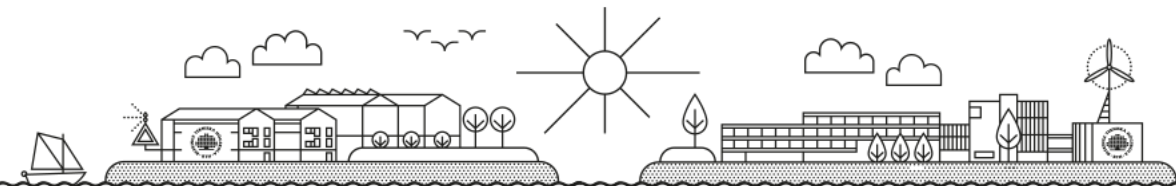


Figure 1.1: The data processing pipeline

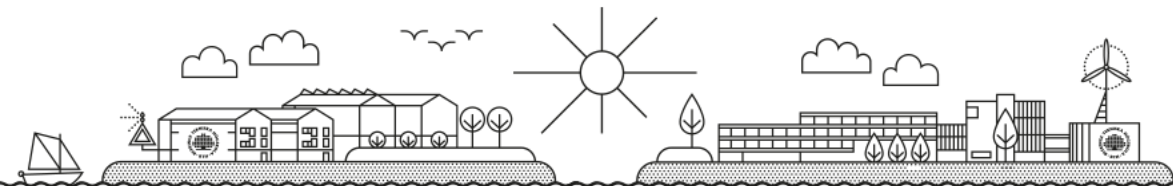
- Data Cleaning and Integration
 - Handle missing and erroneous values
 - Integrate data from multiple sources
- Analytical Processing and Algorithms



PRODUCT RECOMMENDATIONS (1)

*“Example 1.2.1 - Consider a scenario in which a retailer has **Web logs corresponding to customer access to Web pages** at his or her site. Each of these Web pages corresponds to **a product**, and therefore, a customer's **access to a page** may often be **indicative of interest in that particular product**. The retailer also stores **demographic profiles** for different customers.*

*The retailer wants to make **targeted product recommendations** to customers using the **customer demographics** and **buying behavior**.”*



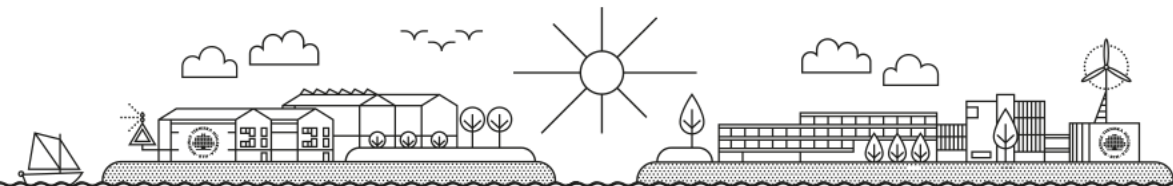
PRODUCT RECOMMENDATIONS (2)

1. Data Collection

- Web logs at the site

```
98.206.207.157 - - [31/Jul/2013:18:09:38 -0700] "GET  
/productA.htm HTTP/1.1" 200 328177 "-" "Mozilla/5.0 (Mac  
OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0  
Mobile/10B329 Safari/8536.25" "retailer.net"
```

- Demographic information within the retailer database



PRODUCT RECOMMENDATIONS (3)

2. Feature Extraction

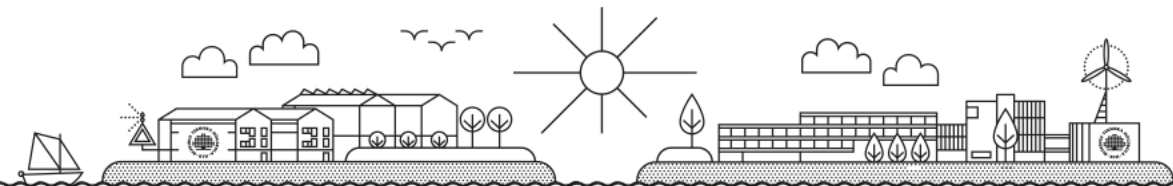
- A specific choice of features extracted from the Web page access
 - E.g., a feature to consider **the number of accesses** to each product description
- Add demographics information

3. Data Cleaning

- Estimate, Remove, Normalization

4. Making Recommendation

- Partition customers by clustering (building block)
- Recommend based on behaviors of customers in the same group
 - For a given customer, the **most frequent items** accessed by the customers in that group are recommended.



BASIC DATA TYPES

1. Nondependency-oriented Data

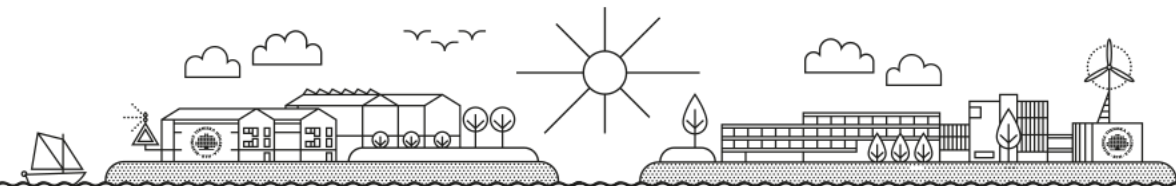
- Data records do not have any specified dependencies between either the data items or the attributes

Table 1.1 An example of a multidimensional data set

Name	Age	Gender	Race	ZIP code
John S.	45	M	African American	05139
Manyona L.	31	F	Native American	10598

2. Dependency-oriented Data

- Implicit or explicit relationships may exist between data items
 - Social Network (**vertices**: data items, **edges**: relationship),
 - Time Series (successive values collected from a sensor)

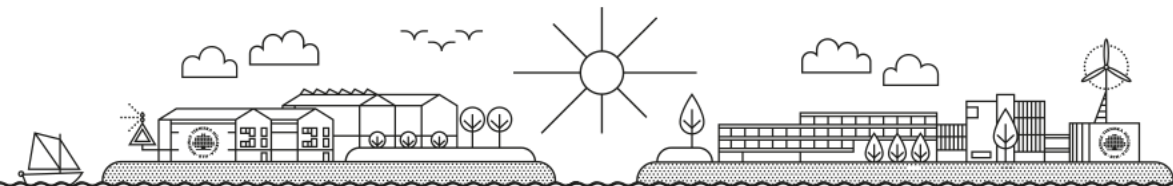


NONDEPENDENCY-ORIENTED DATA (1)

○ Multidimensional Data (Vectors)

“Definition 1.3.1 A multidimensional data set D is a set of n records, $\overline{X}_1 \dots \overline{X}_n$, such that each record $\overline{X}_i$ contains a set of d features denoted by $\{x_i^1, \dots, x_i^d\}$.”

- *Row*: Record, data point, instance, example, transaction, entity, tuple, object, feature-vector
- *Column*: Fields, attributes, dimensions, features.



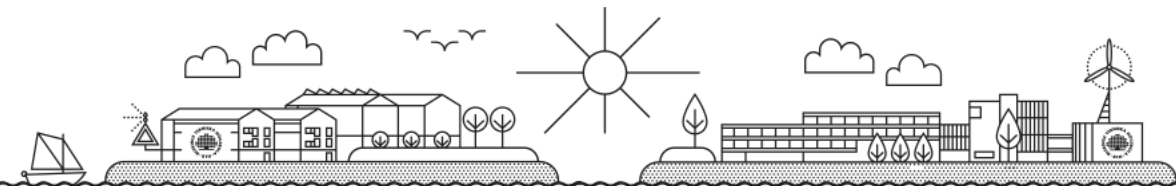
NONDEPENDENCY-ORIENTED DATA (2)

○ Quantitative Multidimensional Data

Table 1.1 An example of a multidimensional data set

Name	Age	Gender	Race	ZIP code
John S.	45	M	African American	05139
Manyona L	31	F	Native American	10598

- Numerical in the sense that they have a natural ordering
- Continuous, numeric, or quantitative
- Convenient for analytical processing
 - Mean, Variance, +/-*/



NONDEPENDENCY-ORIENTED DATA (3)

○ Categorical Data

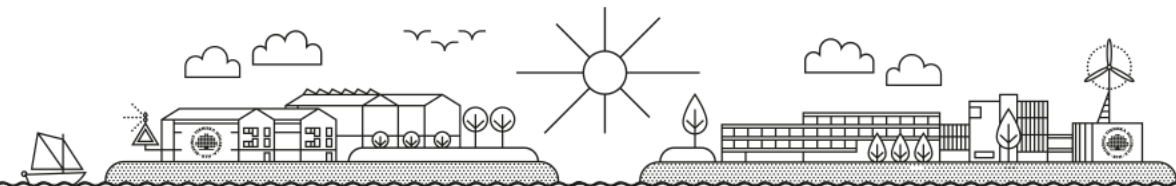
Table 1.1 An example of a multidimensional data set

Name	Age	Gender	Race	ZIP code
John S.	45	M	African American	05139
Manyona L	31	F	Native American	10598

- Take on discrete unordered values

○ Mixed Attribute Data

- A combination of categorical and numeric attributes



NONDEPENDENCY-ORIENTED DATA (4)

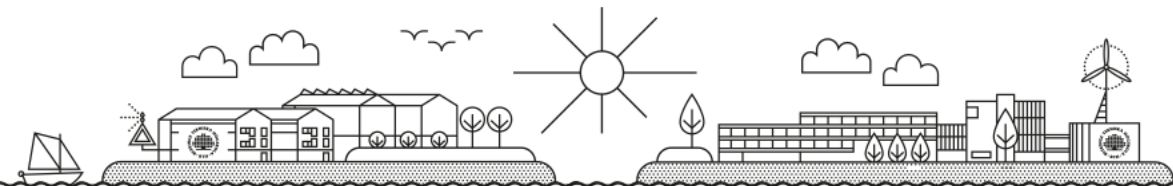
○ Binary Data

- A special case of multidimensional categorical data
 - ✓ Each categorical attribute may take on one of at most two discrete values
- A special case of multidimensional quantitative data
 - ✓ An ordering exists between the two values

○ Setwise Data

- Each attribute is a set element indicator

$$I(x) = \begin{cases} 1, & \text{if } x \in X. \\ 0, & \text{otherwise} \end{cases}$$

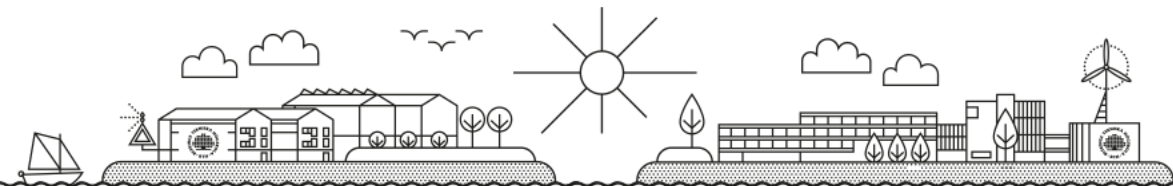


NONDEPENDENCY-ORIENTED DATA (5)

○ Text Data

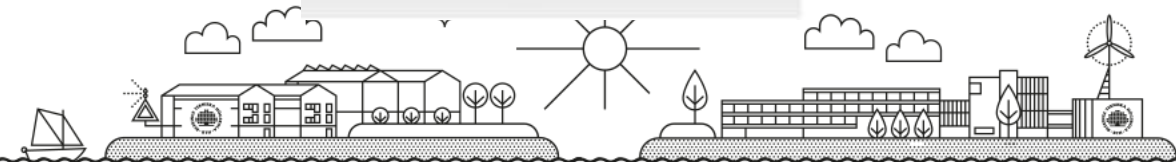
- String—a dependency-oriented data
 - Natural Language Processing
- Document-term matrix—a multidimensional quantitative data

$$\begin{array}{c} D_1 \\ D_2 \\ \dots \end{array} \begin{array}{ccc} w_1 & w_2 & \dots \\ \left[\begin{array}{ccc} 4 & 0 & \dots \\ 2 & 1 & \dots \\ \dots & \dots & \dots \end{array} \right] & m \times n \end{array}$$



DEPENDENCY-ORIENTED DATA (1)

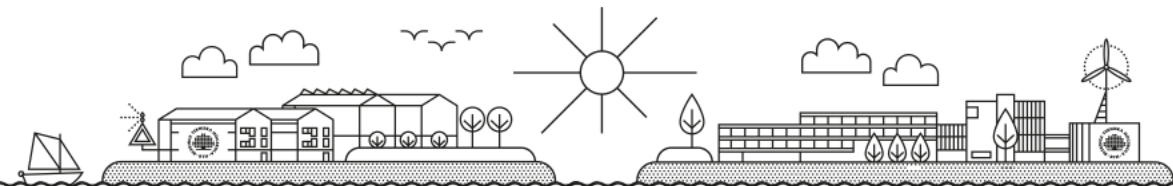
- Implicit dependencies
 - Dependencies are known to “typically” exist
 - E.g., Consecutive temperature values collected by a sensor are likely to be extremely similar to one another.
- Explicit dependencies
 - Graph or network data where edges are used to specify relationships



DEPENDENCY-ORIENTED DATA (2)

○ Time-Series Data

- Contextual attributes
 - Define the context based on which the implicit dependencies occur in the data
 - E.g., time stamp, position index
- Behavioral attributes
 - Represent the values that are measured in a particular context
 - E.g., temperature sensor: measured outdoor temperature
- multidimensional time-series dataset
 - multiple sensors record readings at synchronized time stamps



DEPENDENCY-ORIENTED DATA (3)

○ Discrete Sequences and Strings

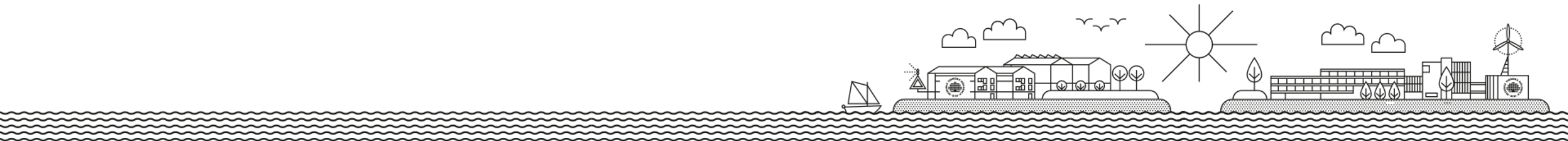
- The categorical analog of time-series data

- Event logs: a sequence of user actions

Login Password Login Password Login Password ...

- Biological data: strings of nucleotides or amino acids

order of such units provides information about the characteristics of protein function



DEPENDENCY-ORIENTED DATA (4)

- Spatial Data

- many nonspatial attributes (e.g., temperature, pressure, image pixel color intensity) are measured at spatial locations.

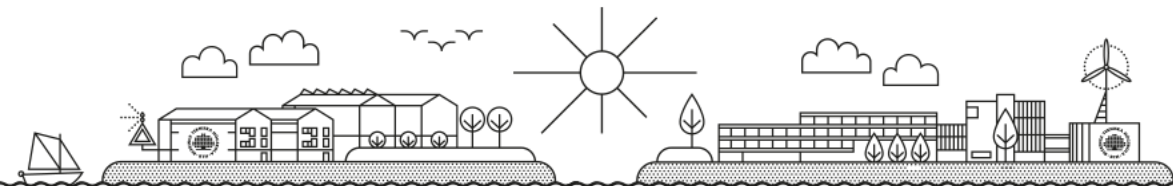
E.g., sea-surface temperature

- Spatiotemporal Data

- Contains both spatial and temporal attributes

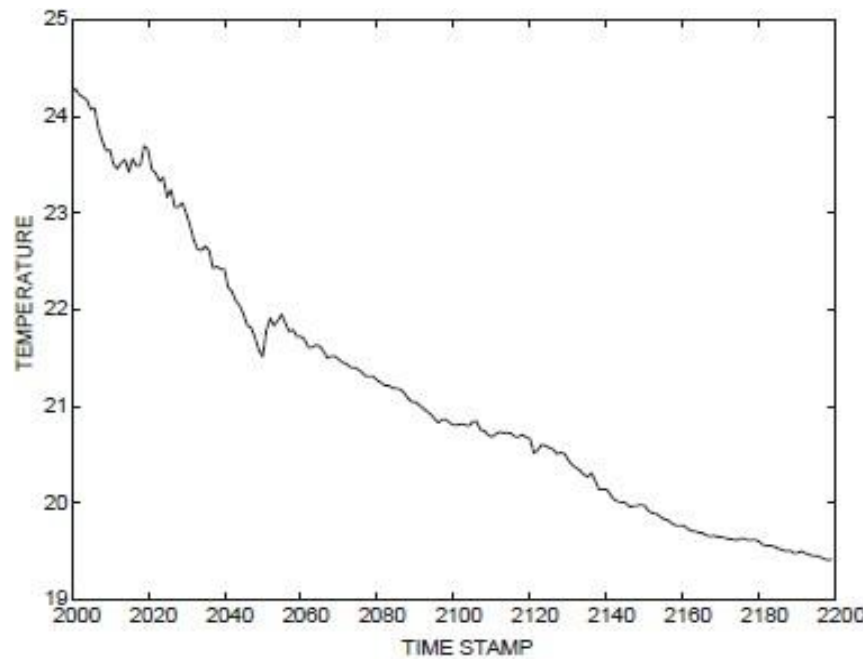
- One of the following can be considered:

1. Both **spatial** and **temporal** attributes are contextual
 - ✓ sea-surface temperature
2. The temporal attribute is contextual, whereas the spatial attributes are behavioral
 - ✓ Trajectory analysis

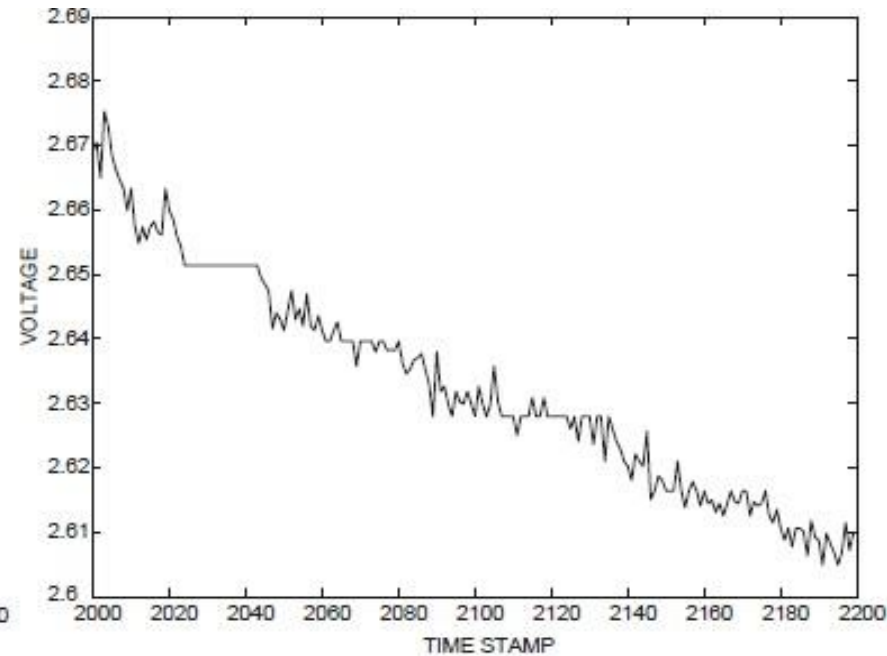


DEPENDENCY-ORIENTED DATA (5)

- Note that any 2- or 3-dimensional time-series data can be mapped onto trajectories.



(a) Temperature

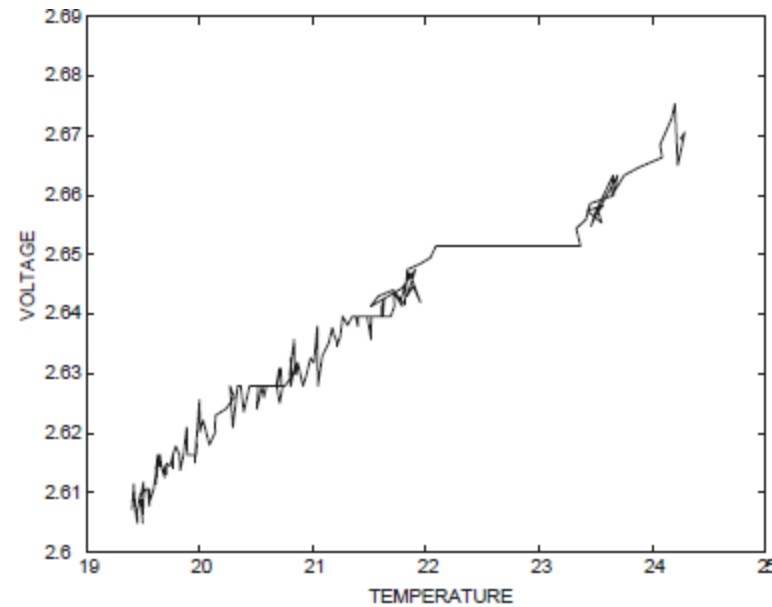


(b) Voltage

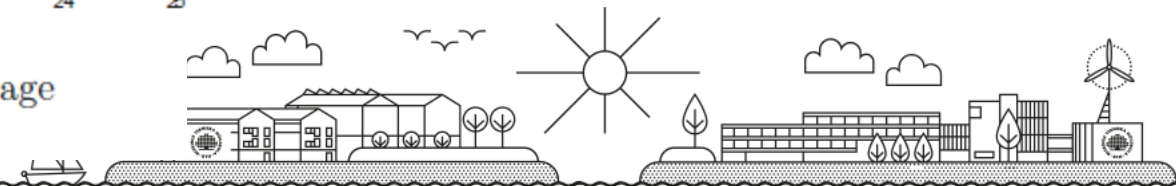


DEPENDENCY-ORIENTED DATA (6)

- Note that any 2- or 3-dimensional time-series data can be mapped onto trajectories.



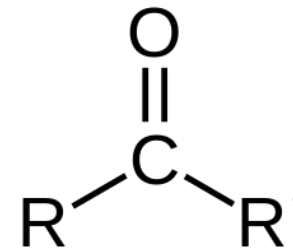
(c) Temperature-voltage trajectory



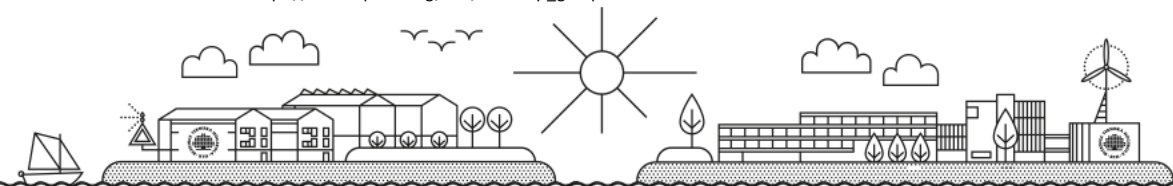
DEPENDENCY-ORIENTED DATA (7)

○ Network and Graph Data

- A single network
 - Web graph with directed edges corresponding to directions of hyperlinks
 - Social networks with undirected edges corresponding to friendships
 - ✓ Emails -> the edges may have content associated with them.
- A database containing many small graphs
 - Chemical compound databases



https://en.wikipedia.org/wiki/Carbonyl_group

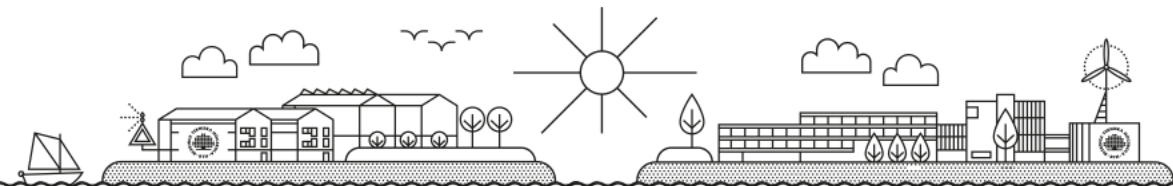


THE MAJOR BUILDING BLOCKS (1)

- Consider a multidimensional database D with n records and d attributes

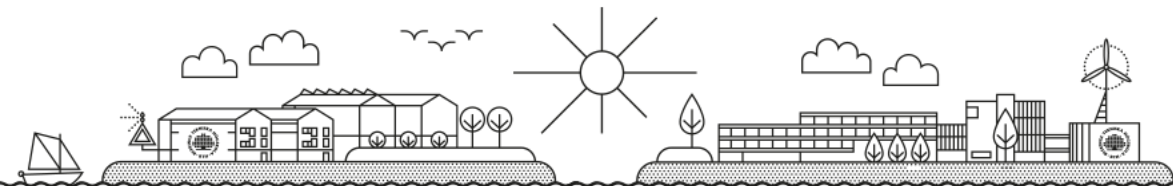
$$\begin{array}{c} \text{d columns/attributes} \\ \begin{bmatrix} 1 & 0 & \dots & 2 \\ 0 & 3 & & 5 \\ 3 & 0 & & 8 \\ 7 & 0 & \ddots & 1 \\ \vdots & \vdots & & \vdots \\ 0 & 2 & \dots & 1 \end{bmatrix} \end{array} \in \text{iff } n \times d$$

n rows/records



THE MAJOR BUILDING BLOCKS (2)

- Consider a multidimensional database D with n records and d attributes
 - Relationships between columns
 - ✓ Positive or negative association pattern mining problem
 - ✓ Data classification (i.e., Prediction)
 - Relationships between rows
 - ✓ Clustering
 - ✓ Outlier analysis

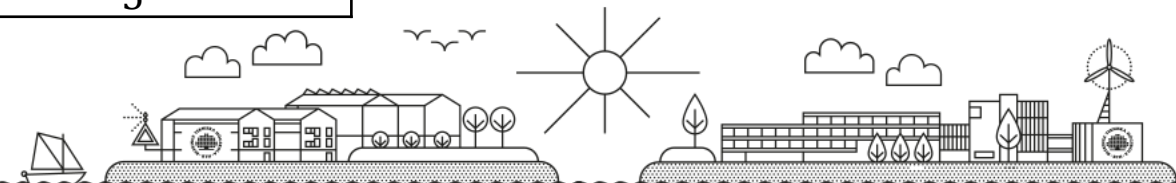


ASSOCIATION PATTERN MINING (1)

- Sparse binary databases

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	
1	0	1	1	0	$\in \{0,1\}^{S \times 4}$
2	1	0	0	1	
3	0	1	1	0	
4	0	1	1	0	
5	1	0	0	1	

Minimum support	Frequent Items	Support
$\frac{2}{5}$	{a, d}	$\frac{2}{5}$
	{b, c}	$\frac{3}{5}$



ASSOCIATION PATTERN MINING (2)

○ Association Rule Mining

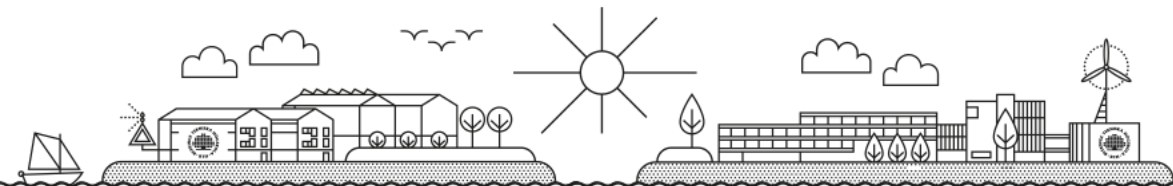
- Rule $A \Rightarrow B$

- If A **appears**, then B also **appears**

- The **confidence** of the rule

$$\frac{\text{support}(A \cup B)}{\text{support}(A)}$$

support($A \cup B$): support of the patterns A and B, i.e., the number of times items A and B appear **together**.

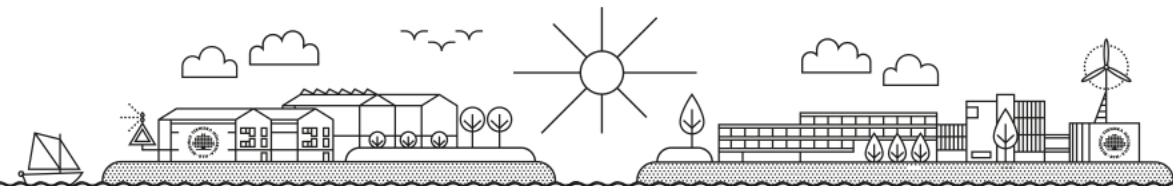


DATA CLUSTERING (1)

“Definition 1.4.3 (Data Clustering) Given a data matrix D (database D), partition its rows (records) into sets $C_1 \dots C_k$, such that the rows (records) in each cluster are “similar” to one another.”

○ Important aspects:

- How to measure the similarity?
 - Human (Nationality, Gender, Age)
- What is the number of clusters?
- Do clusters overlap with each other?
- How to measure the quality of a partition?

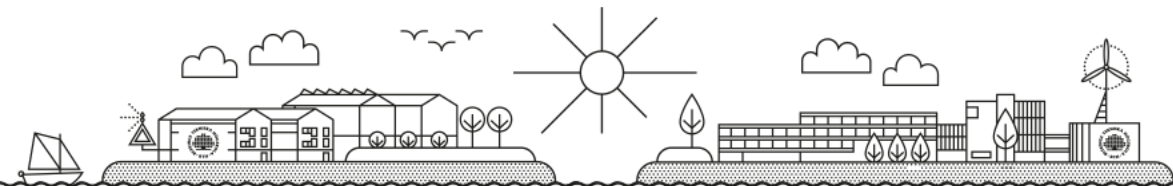


DATA CLUSTERING (2)

“Definition 1.4.3 (Data Clustering) Given a data matrix D (database D), partition its rows (records) into sets $C_1 \dots C_k$, such that the rows (records) in each cluster are “similar” to one another.”

○ Relevant Applications

- Customer segmentation
 - determine customers that are similar to one another in the context of a variety of product promotion tasks
- Data summarization
 - Identifying representative points
- Application to other data mining problems
 - Unsupervised version of classification
 - Outlier analysis

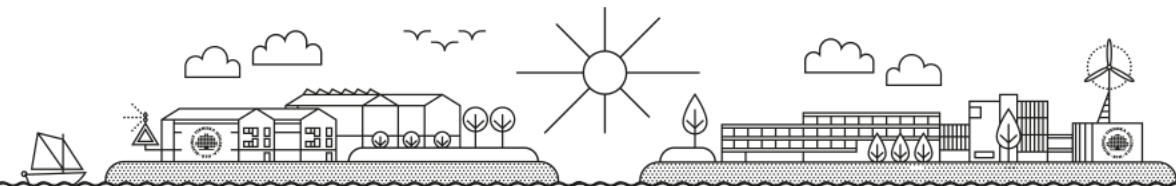


OUTLIER DETECTION

Hawkins:

“An **outlier** is an observation that **deviates so much** from the **other observations** as to arouse suspicions that it was generated by a different mechanism.”

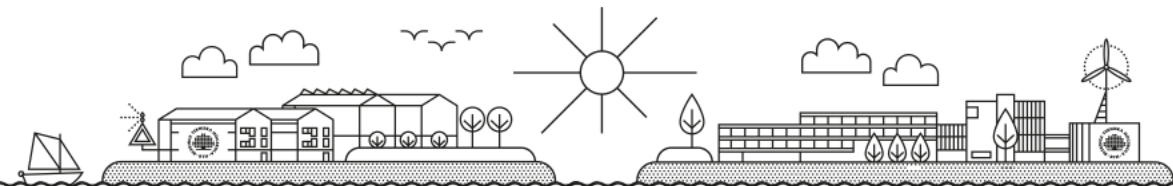
- Abnormalities, Discordants, Deviants, or Anomalies
- Applications
 - Intrusion-detection systems
 - Credit card fraud
 - Interesting sensor events
 - Earth science
 - Medical diagnosis
 - Law enforcement



DATA CLASSIFICATION (2)

“Definition 1.4.5 (Data Classification) Given an $n \times d$ training data matrix D (database D), and a class label value in $\{1 \dots k\}$ associated with each of the n rows in D (records in D), create a training model $\mathcal{M}$, which can be used to predict the class label of a d -dimensional record $Y \notin D$.”

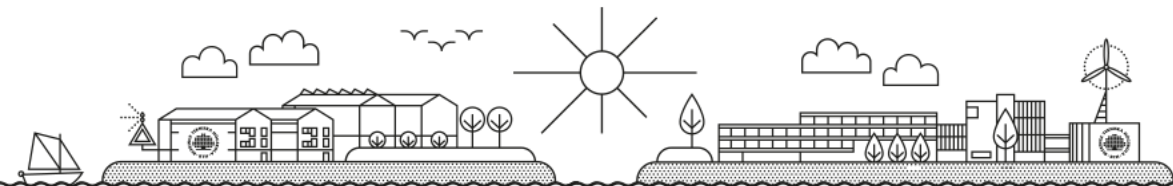
- Relation to Clustering
 - Supervised vs Unsupervised
- Relation to Association Pattern Mining
 - Classification based on association rules
- Relation to Outlier Detection
 - Supervised outlier detection can be modeled as a classification problem



DATA CLASSIFICATION (3)

“Definition 1.4.5 (Data Classification) Given an $n \times d$ training data matrix D (database D), and a class label value in $\{1 \dots k\}$ associated with each of the n rows in D (records in D), create a training model $\mathcal{M}$, which can be used to predict the class label of a d -dimensional record $Y \notin D$.”

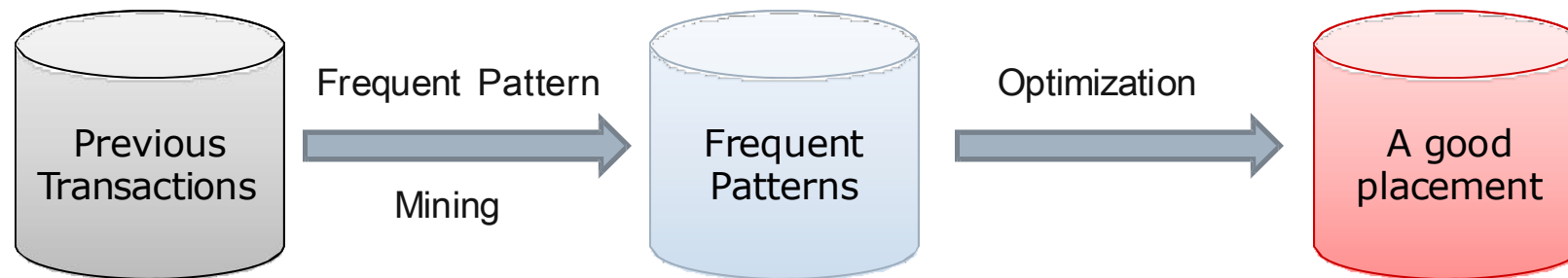
- Applications
 - Target marketing
 - ✓ Predict buying behaviors
 - Intrusion detection
 - ✓ Predict the possibility of intrusions
 - Supervised anomaly detection
 - ✓ Identify records belonging to rare class



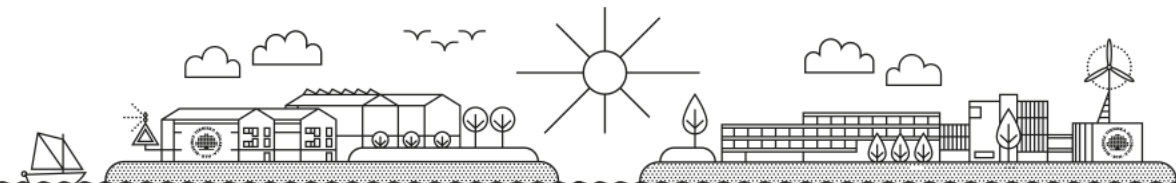
STORE PRODUCT PLACEMENT

- The famous “beer and diapers” story

“Application 1.6.2 (Product Recommendations) A merchant has an $n \times d$ binary matrix D representing the buying behavior of n customers across d items. It is assumed that the matrix is sparse, and therefore each customer may have bought only a few items. It is desirable to use the product associations to make recommendations to customers.”



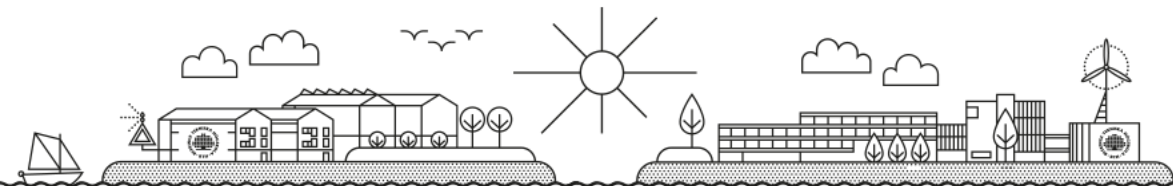
- For each placement, define a score based on frequent patterns



CUSTOMER RECOMMENDATIONS (1)

“Application 1.6.2 (Product Recommendations) A merchant has an $n \times d$ binary matrix D representing the buying behavior of n customers across d items. It is assumed that the matrix is sparse, and therefore each customer may have bought only a few items. It is desirable to use the product associations to make recommendations to customers.”

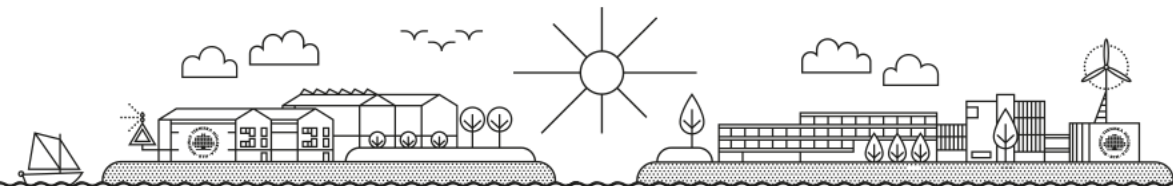
- A simple solution based on association rule mining
 - Find associate rules at particular levels of support and confidence
$$A \Rightarrow B$$
 - If a customer have bought items in A , then it is likely they will buy items in B .



CUSTOMER RECOMMENDATIONS (2)

“Application 1.6.2 (Product Recommendations) A merchant has an $n \times d$ binary matrix D representing the buying behavior of n customers across d items. It is assumed that the matrix is sparse, and therefore each customer may have bought only a few items. It is desirable to use the product associations to make recommendations to customers.”

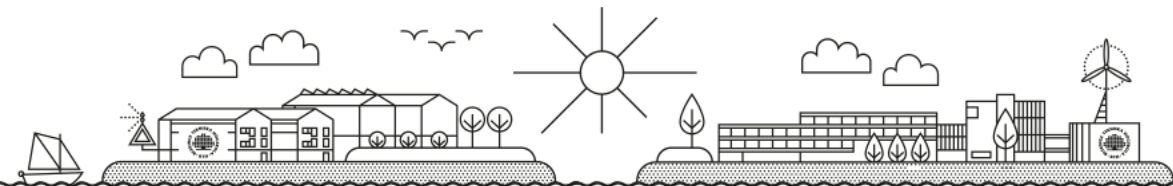
- A second solution based on clustering
 - For a customer, find the most similar customers
 - Recommendation based on items bought by customers similar to they



CUSTOMER RECOMMENDATIONS (3)

“Application 1.6.2 (Product Recommendations) A merchant has an $n \times d$ binary matrix D representing the buying behavior of n customers across d items. It is assumed that the matrix is sparse, and therefore each customer may have bought only a few items. It is desirable to use the product associations to make recommendations to customers.”

- A hybrid approach
 - Apply clustering to partitioning customers to similar groups
 - In each group, use association pattern mining to make recommendations



SUMMARY

- The Data Mining Process
 - Collection -> Proprocessing -> Analytical
- The Basic Data Types
 - Nondependency-Oriented Data
 - Dependency-Oriented Data
- The Major Building Blocks
 - Association Pattern Mining
 - Data Clustering
 - Outlier Detection
 - Data Classification

